Author
Name : Rishi Bhatt
Roll Number : 21f1000330
Email ID : 21f1000330@ds.study.iitm.ac.in

Introduction to LMSv2 project

I have named my library management system as "Pustakalay", because this is the traditional Indian name for a library. In modern world, it is important to have a web-based interface and my "Pustakalay" provides the same using VueJS3 in the front-end, flask based APIs, SQLite3 for DB in the backend supported by redis for caching and celery for backend jobs.

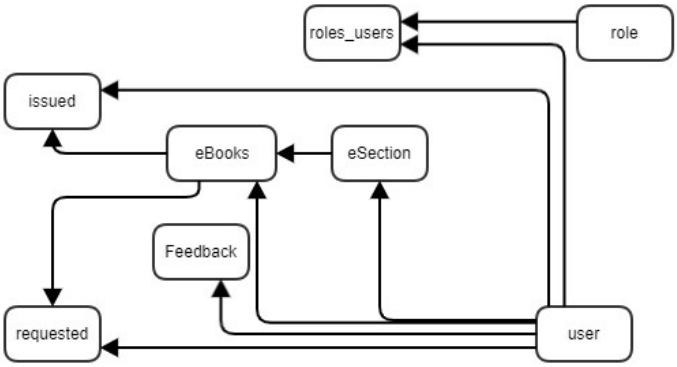

Figure 1: Table structure in SQLite3 database

Figure 1 shows the table structure in SQLite3 database of my project. Relevant foreign keys are used to enable cross-table references wherever needed in the code.

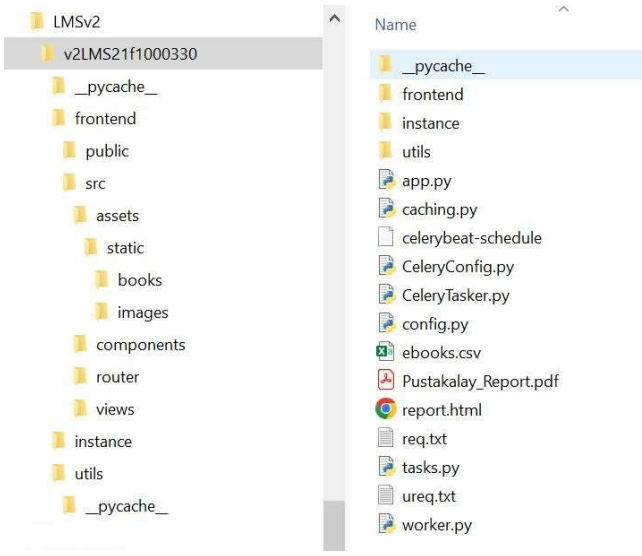

Figure 2: Project Structure

Figure 2 shows the overall project structure.

Libraries and Packages

- Flask: A micro web framework for Python used for backend development.
- Flask-Security-Too: Simplifies the implementation of tokenization, authentication, and authorization.

- SQL Alchemy: A SQL toolkit and Object-Relational Mapping (ORM) library for Python.
- Flask-RESTful : Adds support for quickly building API's
- Flask-CORS : Allows us to handle Cross-Origin Resource Sharing
- Httplib2 : Comprehensive http client library that provides methods for sending http requests and handling responses.
- Celery: A distributed task queue used for asynchronous processing.
- Redis: An open-source, in-memory data structure store used as a database, cache, and message broker.
- Flask-Mail: Provides simple interface for sending emails from our flask application.
- Flask-Caching: Extension that adds caching capabilities to our flask application.
- PyPDF2: A library for reading and manipulating PDF files in Python.
- PIL (Python Imaging Library): A library for image processing tasks in Python
- Vue.js 3: A progressive JavaScript framework used for building user interfaces.
- Vue Router: Vue.js plugin for routing.
- VueX: Vue.js state management pattern and library.
- Axios: Promise-based HTTP client for the browser and Node.js.
- SweetAlert2 : Responsive, customizable,- beautiful, accessible replacement for JavaScript pop-up boxes
- Luxon : Powerful, modern JavaScript Library for working with dates and times.

Challenges Faced

- Token Management :

  Handling token generation, and using the local storage for authentication and authorization of users and administrator, proved to be complex. Ensuring that tokens are securely stored on the client side(in Vue.js) and properly validating it on the server side (in flask) required careful planning and implementation.

- Celery configuration :

  Managing and scheduling periodic tasks in Celery such as the Daily Reminder, Monthly Report, configuring redis with celery, and integrating celery beat schedule, required a deep understanding of Celery's scheduling capabilities and how to properly configure periodic tasks to run reliably. Ensuring Celery works seamlessly with other flask extensions such as flask-mail for sending emails to Mail Hog server and integrating database interactions for sending mails, involves resolving dependency issues and compatibility across various components.

Conclusion

Pustakalay stands as a testament to overcoming significant challenges in developing a modern library management system. Through meticulous integration of a Flask backend and a Vue.js frontend, it offers a seamless user experience that simplifies library operations. By addressing scalability, data management, and user experience design, **Pustakalay** emerges as a robust, efficient, reliable, and adaptable solution for libraries of all sizes, supporting modernization as technology evolves.

Video Link

https://drive.google.com/file/d/1LiwNjbI2wER9TG9LzlnhivJ4QMLeERiF/view?usp=sharing